Nouman Ahmed (He/Him)

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Education

University of Tulsa — Tulsa, OK
Ph.D. Computer Science — expected 2028
Relevant Coursework: Machine Learning, High Performance Computing

University of Engineering and Technology, Lahore — Lahore, Pakistan
B.S. Electrical Engineering; Best final Year Project — June 2021
PEEF Scholar

Research Experience

University of Tulsa — Tulsa, OK
PhD Researcher — July 2025 - present

- Designed evaluation framework for Key-Value cache eviction strategies and found that their rankings didn't transfer to different LLM architectures.
- Developed course content, assignments, and laboratory modules for Artificial Intelligence: Principles & Practices, a high school-level AI curriculum delivered to students in Tulsa, OK.
- Built an NLP pipeline using Sentence Transformers and permutation testing to analyze 2,280 NICE Cybersecurity Framework statements, statistically validating core competencies as the corpus's semantic center ($p < 10^{-4}$) and using them to derive learning outcomes for an Introduction to Cybersecurity course.
- Engineered and developed LLM-based framework to generate over 100 attack graphs from NVD and CVE vulnerability datasets, reducing manual analysis time by 40%.
- Applied statistical methods (MCA & PCA) to analyze pose estimation data for 300+ ancient sculptures and organized them into clusters.

University of Tulsa — Tulsa, OK
Master's Researcher — Sep. 2024 - May 2025

- Designed and constructed an LLM-based AI recommender system for 2,000 users, increasing recommendation diversity by 15%.
- Modeled and implemented a system to sustain cooperation in multi-agent systems using Reinforcement Learning.

University of Engineering and Technology — Lahore, Pakistan
Undergraduate Researcher — June - Aug. 2021

- Designed and built a real-time fault detection and monitoring prototype for power systems, achieving <1 second latency in reporting and 95% accuracy in fault identification.

Leadership Experience

University of Tulsa — Tulsa, OK
Teaching Fellow — Sep. 2023- present

- Taught and assisted in four undergraduate courses with class sizes ranging from 9 to 280 students.
- Instructed students on topics including Introduction to Programming, Fundamentals of Algorithms, and Database Systems.
- Prepared comprehensive course materials, including laboratory experiments, lectures, exams, and homework.
- Led weekly laboratory and discussion sections for groups of 13-30 students.
- Supervised student final projects and evaluated exams and weekly homework.

University of Tulsa Tulsa, OK
Graduate Student Association (GSA) Senator June 2025 - Present

- Represented the Computer Science department and voted on graduate travel awards and organizational funding requests.
- Served on the Public Relations Committee.

University of Tulsa Tulsa, OK
Secretary in Machine Learning and AI Club August 2026

- Maintain accurate participation, member, and breakout group information.
- Maintain informational availability for all members of the club in communication and file-sharing platforms such as MS Teams, OneNote, Google Drive, Discord, GitHub, or others.
- Ensure the security and availability of admin-level information, records, etc.
- Find and present events that bolster the community participation of the club.
- Coordinate with Directors of Marketing to contact and notify members about meetings and events.

Skills

Computer Languages: Python, Java, C#, C++, SQL, JavaScript, TypeScript
Machine Learning Frameworks: PyTorch, TensorFlow, JAX, scikit-learn, Hugging Face
Domains: NLP (Transformers, LangChain), Computer Vision (OpenCV), Reinforcement Learning (Gymnasium, RLlib)
MLOps: MLflow, W&B, DVC, Docker, AWS SageMaker, FastAPI

Publications and Submissions

Published

"Analysis of Pose Estimation of West Mexican Ceramic Figures using Machine Learning." EVA London Conference Proceedings. 2026.
"KV-Cache Eviction Rankings Need Not Transfer: A Paired Comparison on Qwen3-8B and Llama-3.1-8B". 2026 International Conference on Large Language Models (LLM)

In Preparation

"Pose Estimation of West Mexican Ceramic Figures: A Machine Learning Landscape." Journal of Archaeological Science. (Manuscript in Submission)
"An Analytical Approach to Cybersecurity Curriculum Design."